

AI-Generated Image Detection

Project Proposal — CSE 429: Computer Vision & Pattern Recognition

Egypt-Japan University of Science and Technology (E-JUST) | Spring 2026 | Dr. Ahmed Gomaa

Course	CSE 429	Semester	Spring 2026
Instructor	Dr. Ahmed Gomaa	Group Size	6 Members
Type	Review & Implement	Institution	E-JUST

Team Members

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1. Problem Statement

Given any input image, output a binary prediction—**Real** or **Fake (AI-Generated)**—along with a confidence score. The system must generalise to generator families not seen during training.

The rapid proliferation of AI-generated images from GANs and Latent Diffusion Models has created an urgent need for reliable detection systems. The core challenge is **cross-generator generalisation**: a detector trained on one generator family typically fails on unseen generators.

- **Input:** Any RGB image (photograph or AI-generated)
- **Output:** Binary label (Real / Fake) + confidence score $\in [0, 1]$
- **Challenge:** Generalise across generator families not seen during training

2. Approach

We re-implement **UniversalFakeDetect** (Ojha et al., CVPR 2023), which classifies images in the feature space of a frozen CLIP-ViT-L/14 encoder—trained on 400M image-text pairs with no prior exposure to the real-vs-fake distinction.

Backbone	Frozen CLIP-ViT-L/14 (307M parameters, zero gradient updates)
Head	Single trainable Linear(768→1) — only 769 parameters trained
Loss	Binary cross-entropy with logits (BCEWithLogitsLoss)
Optimiser	AdamW (lr=0.001, weight decay=0.001), Cosine Annealing LR
Hardware	NVIDIA RTX 3050 6GB Laptop GPU
Deployment	Gradio web application on Hugging Face Spaces

The key insight: CLIP features capture universal visual semantics without generator-specific fingerprints, avoiding the “sink class phenomenon” that causes prior methods to fail on unseen generators.

3. Experiments & Results Plan

Datasets	CIFAKE (Stable Diffusion v1.4) + Real & Fake Face Detection (StyleGAN)
Split	70% train / 15% val / 15% test (stratified random split, seed=42)
Baselines	Random Classifier + Simple 4-layer CNN trained from scratch
Metrics	AUC-ROC, Accuracy, F1-Score, Average Precision (AP)
Success	AUC > 0.90 on the combined test set

Key References

- [1] U. Ojha, Y. Li, Y. J. Lee, *Towards Universal Fake Image Detectors that Generalize Across Generative Models*, CVPR 2023.
- [2] A. Radford et al., *Learning Transferable Visual Models From Natural Language Supervision*, ICML 2021.
- [3] J. J. Bird, A. Lotfi, *CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images*, arXiv:2303.14126, 2023.